

**Case
Study**

Cropin

**Enabling Efficient Farm Operations
For One Of The World's Largest
Agrochemical Corporation Through
Satellite Imaging**

THE CLIENT

The client is an American agrochemical and agricultural biotechnology corporation founded in 1901 with headquarters in St. Louis, Missouri, United States. The organization is one of the top 10 US chemical companies and one of the first companies to apply the biotechnology

industry business model to agriculture, using techniques developed by biotech drug companies. The company was ranked within the first 200 on the 2018 Fortune 500 of the largest United States corporations by revenue.

THE CHALLENGE

Monitoring of season-wise maize acreage at different stages held significance in sound marketing planning and inventory management for the client. The organization had no insights on the maize acreage at various scales from village level to district level and acreage estimation was carried out by the field staff using basic methods like crop cuts, farmer survey, harvest sample method, whole plot harvest, expert assessments, crop cards, and allometric methods, which led to data inconsistencies quite often.

Some of the major challenges faced by the organisation :



Manual Intervention in Crop Acreage Estimation

- Labour intensive process
- Cost ineffective for large-scale estimation
- Inadequate monitoring due to lack of standard benchmarks



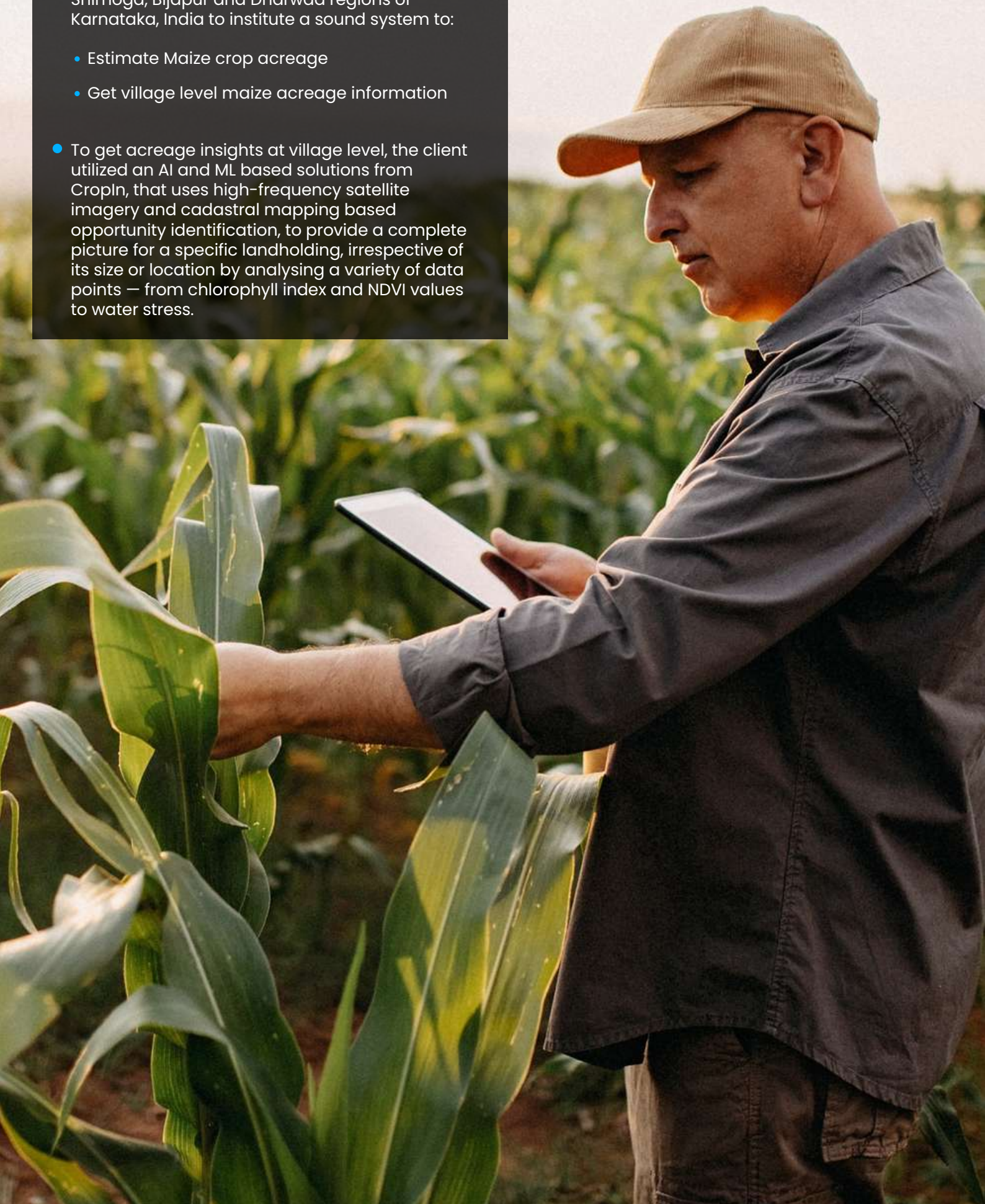
Data Quality Issues

- Error-prone data acquisition methods.
- Inaccurate reporting of yield at the plot level.
- Poor data verification at different levels of management.

Cropln's Innovation



- The client partnered with Cropln in 2019 to implement its technology solutions in Bellary, Shimoga, Bijapur and Dharwad regions of Karnataka, India to institute a sound system to:
 - Estimate Maize crop acreage
 - Get village level maize acreage information
- To get acreage insights at village level, the client utilized an AI and ML based solutions from Cropln, that uses high-frequency satellite imagery and cadastral mapping based opportunity identification, to provide a complete picture for a specific landholding, irrespective of its size or location by analysing a variety of data points — from chlorophyll index and NDVI values to water stress.





RESULTS



- CropIn's successful implementation of its digital technologies enabled the client to remotely monitor village wise Maize acreage of last three years across Bellary, Shimoga, Bijapur and Dharwad districts of Karnataka, India
- CropIn leveraged AI and ML-based satellite imagery to estimate acreage based on the crop's water requirement with the help of remote-sensing derived phenology parameters, environmental conditions, crop yield specific coefficients and data collected on plot history, existing vegetation and sowing window.
- Village level acreage identification for maize allowed the company to identify the villages to concentrate for internal sales and inventory management along with deciding on marketing strategy for the upcoming seasons enabling better management of delinquencies.
- It can be observed, the combination of convenience and ease of use, along with a smart package of numerous practical yet unique features has resulted in the success of the CropIn - client partnership. In the years to come, CropIn is set to scale up its services by leveraging its cutting-edge AI- and ML-based solutions for all-round progress of the client's operations.

State	District	Season	Crop	2017 (Ha.)	2018 (Ha.)	2019 (Ha.)
Karnataka	Bellary	Kharif	Maize	115381	131114	105441
Karnataka	Bijapur	Kharif	Maize	62330	46616	73659
Karnataka	Dharwad	Kharif	Maize	26853	31415	14374
Karnataka	Shimoga	Kharif	Maize	56023	70745	61859

2017, 2018 and 2019 Maize Acreage results of 4 districts



Figure. a.b. and c.are 2017, 2018 and 2019 (Ha) Maize acreage of Bellary



Figure. d.e. and f. are 2017, 2018 and 2019 (Ha) Maize acreage of Dharwad



Figure.g.h. and k. are 2017, 2018 and 2019 (Ha) Maize acreage of Bijapur



Figure.g.h. and k. are 2017, 2018 and 2019 (Ha) Maize acreage of Shimoga



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